MATH 180 Strategies of Problem Solving

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Outline

1. SoPS Lecture 15
   - Ready Set Bet
   - Expectation Value
   - Wrapping up

Review

Last time

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- Stars and bars

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Last time

- Stars and bars
- working on Connect 4 problems

Outline

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Play the game

Activity

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Activity

- play a couple rounds of Ready Set Bet

Play the game

Activity

- play a couple rounds of Ready Set Bet
- what strategies seem to be the best?

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Starting example

Problem

Your friend proposes the following bet

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- roll two 6-sided die

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- gain a dollar if we roll non-double

Starting example

Problem

Your friend proposes the following bet

- roll two 6-sided die
- gain a dollar if we roll non-double
- lose 8 dollars if we roll doubles

Should you play?

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- the set Ω of all possible outcomes is the **sample space**
- subsets $E \subseteq \Omega$ are called **events**
- the **probability** of an event E is

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|}$$

Back to example

The **sample space** is:

$$\Omega = \{(1,1),(1,2),(1,3),\ldots,(6,6)\}$$

The **event** of rolling doubles is:

$$E = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}$$

The probability of rolling doubles is

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{6}{36} = \frac{1}{6}$$

The probability of NOT rolling doubles is

$$\mathbb{P}(E) = 1 - \frac{1}{6} = \frac{5}{6}.$$

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If you take the bet, your probability of winning is about 83%.

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... or should you?

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A **value function** $f : \Omega \to \mathbb{R}$ assigns to each possible experimental outcome, the the value of the outcome.

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$$\mathbb{E}(f) = \frac{\overbrace{1+1+\cdots+1}^{30 \text{ terms}} + \overbrace{(-8)+\cdots+(-8)}^{6 \text{ terms}}}{36} = \frac{30-48}{36} = -\frac{1}{2}$$

Back to example

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Solution

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Each time you take the bet, you lose an average of 50 cents.

Back to example

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Don't take the bet!

Ready Set Bet

Discussion

How can expectation values help us in Ready Set Bet?

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Final thoughts

Reflection

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- Revisit the two dice bet where if you roll non-doubles you win a dollar, but if you roll doubles, you lose m dollars. Which value of m would result in a "fair bet"? Why?

Submit your responses by **Tonight!**